

Functional Requirements

Function 1: Recipe Management

Users can create, browse, update, rate, and delete recipes. Every recipe stores structured nutritional data automatically calculated from its ingredients. Recipes are shared publicly across all platform users.

Actors

Actor	Description
Registered User	Can perform all CRUD operations on their own recipes
Guest User	Can browse and view recipe details (read-only)
System	Automatically calculates total and per-serving nutrition upon save

Functional Requirements — Recipe Management

ID	Feature	Description
FR-1.1	Browse Recipes	Display all recipes with keyword search (partial title, case-insensitive), cuisine type filter, and max-calories-per-serving filter. Cards show title, category, average rating, thumbnail, and calorie count.
FR-1.2	View Recipe Detail	Any user (auth or not) can view title, description, cuisine, category, dietary tags, prep/cook time, servings, ingredients, step-by-step instructions, total/per-serving nutrition, and community rating.
FR-1.3	Create Recipe	Authenticated users create recipes by providing title, category, cuisine, servings, ingredients (name/qty/unit/macros), and steps. System auto-computes totalNutrition and perServing on save.
FR-1.4	Edit Recipe	Only the recipe creator can edit their own recipe. System-seeded recipes (isUserCreated: false) are not editable. Nutrition recalculates on ingredient/serving changes.
FR-1.5	Delete Recipe	Only the recipe creator can permanently delete their recipe. System-seeded recipes cannot be deleted.
FR-1.6	Rate Recipe	Authenticated users assign 1–5 stars to any recipe. One active rating per user per recipe (updates on re-submit). averageRating is recalculated and persisted after each submission.

FR-1.7	Favorites	Users toggle recipes into/out of their personal favorites via POST (toggle) or DELETE (unconditional remove). Favorites are accessible from the profile page.
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Validation Rules — Recipe

Field	Validation Rule
title	Required, non-empty string
category	Must be one of: Breakfast, Lunch, Dinner, Snack
servings	Required, positive integer
ingredients[].name	Required, non-empty string
rating	Integer between 1 and 5 (inclusive)
image	Optional Base64-encoded string; server rejects payloads > 15 MB

Function 2: Weekly Meal Planning

Registered users can assign recipes to specific meal slots (Breakfast, Lunch, Dinner) for each day of any week. The system persists the plan per user and per week start date. From a saved plan, users can generate a consolidated grocery shopping list.

Functional Requirements — Meal Planning

ID	Feature	Description
FR-2.1	Select a Week	Navigate to any week via a date picker. Monday is used as the startDate key. Only one plan per (user, startDate) is allowed (unique DB index).
FR-2.2	Assign Recipe to Slot	Assign any recipe to a day/meal-slot (e.g., Tuesday Lunch). Valid days: monday–sunday. Valid slots: Breakfast, Lunch, Dinner. One recipe per slot; a new assignment replaces the previous.
FR-2.3	Clear a Slot	Remove the recipe from any individual day/slot, resetting it to null/empty.
FR-2.4	Save the Entire Week	Save all slot assignments for the current week in a single bulk PUT /mealplan operation. If no plan exists for the week, one is automatically created (upsert).

FR-2.5	View a Saved Plan	Return a weekly plan with all recipe slots fully populated (title, image, nutrition, etc.), not just IDs.
FR-2.6	Generate Grocery List	Collect all unique recipes across all 7 days × 3 slots. Aggregate ingredient quantities by name (case-insensitive). Return merged list with name, total quantity, unit, recipe titles, and serving counts.

Business Rules

Rule	Description
BR-2.1	startDate is always normalized to UTC midnight to prevent timezone-related duplicates
BR-2.2	One meal plan document per user per week (upsert strategy)

Rule	Description
BR-2.3	Shopping list merges quantities only when ingredient names match exactly (after lowercase trim)
BR-2.4	Snacks slot is supported in the data model but not yet surfaced in the current week view